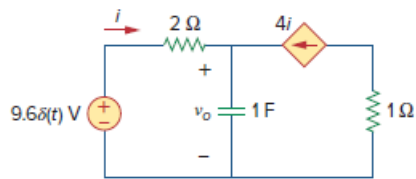


Problem

The capacitor in the circuit of Fig. 16.39 is initially uncharged. Find $v_o(t)$ for $t > 0$.

Figure 16.39:



Step-by-step solution

Step 1 of 4

The circuit initial capacitor current conditions are $i(0^-) = 0$ and $v_o(0^-) = 0$.

Convert the circuit from time domain to s-domain.

Apply the Laplace Transform for the following elements in the circuit

For input voltage element:

$$\begin{aligned} L[9.6\delta(t)] &= 9.6L[\delta(t)] \\ &= 9.6(1) \\ &= 9.6 \text{ Volts} \end{aligned}$$

For capacitor element:

$$\begin{aligned} L\left[\frac{1}{sC}\right] &= L\left[\frac{1}{s \cdot 1}\right] \\ &= \frac{1}{s} \end{aligned}$$

Step 2 of 4

Construct the circuit in s-domain.

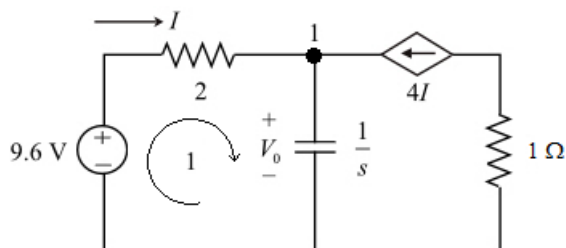


Figure 1

Step 3 of 4

Apply Kirchhoff's current law at node 1.

$$I(s) + 4I(s) = \frac{V_0(s)}{\frac{1}{s}}$$

$$V_0(s) = \frac{5I(s)}{s}$$

$$I(s) = \frac{sV_0(s)}{5} \dots\dots (1)$$

Apply Kirchhoff's voltage law (KVL) equation around loop 1.

$$9.6 - 2I(s) - V_0(s) = 0 \dots\dots (2)$$

Substitute $\frac{sV_0(s)}{5}$ for $I(s)$ in equation (2).

$$9.6 - 2\left(\frac{sV_0(s)}{5}\right) - V_0(s) = 0$$

$$9.6 = \left(\frac{2s}{5} + 1\right)V_0(s)$$

$$V_0(s) = \frac{9.6 \times 5}{2s + 5}$$

$$V_0(s) = \frac{9.6 \times 5}{2 \cdot \left(s + \frac{5}{2}\right)}$$

$$V_0(s) = \frac{24}{\left(s + \frac{5}{2}\right)} \dots\dots (3)$$

Step 4 of 4

Find the Inverse Laplace Transform of $V_0(s) = \frac{24}{\left(s + \frac{5}{2}\right)}$.

Apply Laplace Transform to the equation 3

$$L^{-1}(V_0(s)) = L^{-1}\left(\frac{24}{s + 2.5}\right)$$

$$v_0(t) = 24L^{-1}\left(\frac{1}{(s + 2.5)}\right)$$

$$v_0(t) = 24e^{-2.5t} \text{ Volts}$$

Therefore, the voltage response across the capacitor in an RLC series circuit is:

$$\boxed{v_0(t) = 24e^{-2.5t} \text{ V } t > 0}.$$